

PhyRoute: Voltage Screening under Feeder Reconfiguration

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Abstract. Decision points for a voltage surrogate to go to an exact solver for a changed feeder need to be determined. We investigate a neural approach, a purely physics-based approach as well as a physics-augmented approach to screen voltage feasibility on a synthetic 33-bus test system, with 15 development and 44 held-out branch exchanges, three training seeds and three test seeds. For a frozen threshold calibrated for 10% escalation, the unsafe-approval rate is 0.0065% and the false-rejection rate is 0.0248% for the residual model using current LinDistFlow information. In comparison, the physics-only routing approach has unsafe approvals of 0.0615% and false rejections of 0.1678%. The realized escalation was 8.40% for the residual model and 8.66% for the physics-based routing. However, the residual route has a higher computational cost. A 10% underestimate of the impedance leads to an unsafe rate of 2.24%, stale topology even to 14.86%. Of the 1.83 million generated test cases 2,315 remained numerically unresolved and therefore are not counted in the decision-risk denominators. The results of our investigation show that a learned correction with current network information improves screening under the tested conditions but remains limited by calibration and verifier accuracy.

Keywords: selective prediction, power flow, topology shift, residual learning, calibration, runtime verification

1 Introduction

A voltage surrogate can quickly approve an operating point but then its decisions might be worthless after a change to the feeder. A predictor that has learned on one network does not necessarily receive information that distinguishes two different switch configurations. Even if it is given the same loads, the same amount of PV injection, and the same slack voltage setting, the physical voltage profiles may change. Repeated calls to the predictor with the same inputs may return confident answers even though the physical voltage profile changes.

Exact power-flow fallback does not settle which operating points to verify. Verifying every case would defeat the purpose of using a surrogate for fast approval. We evaluate routing choices and the tradeoffs between verifying too few or too many cases, and between rejecting uncertain cases and possibly missing feasible operation. In particular, we compare a physics-assisted neural route with the same physics model used directly, with exact power-flow fallback.

We apply this study to a synthetic, 33-bus radial test case with balanced power flow, and examine the performance of a number of different models. A ‘stale’ neural

ensemble has access to the same features used for operating point selection (i.e., with no reference to the current topology) while LinDistFlow uses the supplied network model. A second model, a residual predictor, is trained across development topologies using the current LinDistFlow voltage extrema as input, and is then evaluated against physical reference labels on held-out exchanges. This information difference between the models is intentional.

We pair the same operating points across every connected radial single branch exchange in the feeder inventory. We keep development exchanges separate from held-out exchanges, fix thresholds using independent calibration data and perturb the verifier while keeping the physical reference unchanged. We retain unresolved reference solves.

1.1 Related work and scope

A key tradeoff in reject-option prediction is that of error versus refusal [3]. Extending the notion of learning to defer to a downstream expert in a cascading classification architecture was previously studied in [6]. How cascade analysis can explain the confidence of the first model in a cascading architecture of experts for classifying instances into labeled categories, and why the first model’s confidence need not identify when another model in the cascade is actually helpful was previously studied in [4]. Deep ensembles motivate the present member-margin spread diagnostic [5]; it does not reproduce every component of that method. Research on Calibrate-Then-Delegate studies risk and budget guarantees for a cascade of classifiers [9]. The local probes included in this paper do not attempt to provide such guarantees. Instead, this paper includes a split-conformal margin interval for the check decisions and states its exchangeability requirement [1].

This basic architecture of combining a complex learned component with simpler verification has been studied before [10]. For power systems in particular, also feasibility of learned models has been studied before, like in DeepOPF+ [15] where constraints are calibrated in the training process, or in work by Venzke et al. [14] where worst-case guarantees for neural optimal power flow are studied. In work by Parker [8] adversarial AC-power-flow inputs distinguish surrogate feasibility from physical feasibility. We check against the physical governing equations of power systems, not against the confidence of a learned model predicting feasibility.

This work is in no way to be confused with the first use of physics or exact fallback in power-flow studies. The work of Shamseldeen [11] was a combination of a learned model for AC feasibility, a conformal predictor and an exact Newton–Raphson solver. In the work of Nawaz et al. [7] physics-informed graph neural networks were used as surrogates, with a residual-triggered Newton–Raphson fallback and a variety of topology perturbation tests. Here we contrast a number of different routing choices with their required information, and check their performance against physics-only routes, account for their false-rejection rate, and report their computational cost. We test one feeder family, with a set of specified synthetic operating points, and a fixed set of branch additions and removals.

2 Benchmark and routing methods

2.1 Voltage decision and information

The feeder uses the 33-bus data associated with Baran and Wu [2]. The task is a voltage-band decision under a balanced steady-state model, not dispatch optimization or a field-safety claim. For voltage magnitudes v , define

$$m(v) = \min \left\{ \min_i v_i - 0.95, 1.05 - \max_i v_i \right\}. \quad (1)$$

A resolved reference point is feasible when $m(v) \geq 0$. The neural margin m_N uses the ensemble-mean minimum and maximum voltages. The spread σ is the sample standard deviation of the five individual member margins. LinDistFlow returns a margin m_L from approximate voltage drops on the supplied radial network [2, 13]. The 71 features are the 32 active power features, 32 reactive power features, 6 PV features and 1 slack feature (voltage at one bus). These features exclude topology. For the generator, a global load multiplier and per-bus multipliers are sampled from [0.3,1.1], [0.7,1.3] respectively, and the slack voltage is sampled from [0.97,1.05] pu. All six PV outputs are zero together with probability 0.25; otherwise irradiance, plant scaling and retained output are sampled independently within the recorded configuration. An additional test is run for the original topology, with the photovoltaic ratings increased by 60%, and the global load multipliers are sampled from [0.2,0.8].

2.2 Training and topology allocation

Three training seeds each generate 30,000 nominal-topology training points, 10,000 separate probe-training points and 10,000 threshold-calibration points. Each ensemble contains five independently initialized ReLU networks with two 32-unit layers. Networks use Adam, batch size 256, learning rate 10^{-3} and at most 250 epochs. Early stopping uses a 10% internal validation split, patience of 25 epochs and tolerance 10^{-7} . Input and regression-target scalers are fitted on training data. Stopping histories and warnings are retained.

There are 59 connected radial nonbaseline single branch exchanges in the inventory. Fifteen of these are development topologies (including the two from the exploratory study) and the remaining 13 development exchanges follow a fixed hash rule. The remaining 44 exchanges are the primary hold-out set for the experiments in this paper. This protocol was frozen after initial exploratory analysis and was not preregistered before any observation.

A physics-augmented residual network was trained with two 64-unit layers and the same optimizer and stopping settings as the ensemble. The input l contained the minimum and maximum LinDistFlow voltages of an operating point. The corrected prediction for this case was

$$\hat{y}_R(x, l) = l + h_\theta([x, l]). \quad (2)$$

The network fits standardized exact-minus-linear extrema with squared-error regression. Its corrected extrema define margin m_R . For each training seed, this baseline uses 3,000 points from each development topology and 3,000 points from the original topology (for a total of 48,000 points), and does not use the held-out-topology labels in either the fitting or the calibration. It gets current-physics information not encoded in a graph, and reports additional training data for this baseline as cost.

Three independent test seeds, each generating 10,000 operating points paired across the original feeder and all 59 exchanges, and three further nominal-topology “photovoltaic-growth” sets, generating 1,830,000 topology–scenario pairs, are available for reuse in the runs to test the verifiers. As noted above, reuse across training runs and verifier variants does not generate additional independent labels. Hence, there are nine model–test pairs for each of the conditions.

2.3 Routes and frozen thresholds

A route combines a provisional verdict with a score; larger scores receive exact evaluation first. Table 1 defines the twelve score routes. Let $a = 1[m_N \geq 0]$, $b = 1[m_L \geq 0]$ and $c = 1[m_R \geq 0]$. The violation classifier estimates p from the original training set. A separate two-layer, 32-unit probe fits $\log_{10}(|m_N - m| + \epsilon)$ using $[x, m_N, \sigma]$ from the probe split, with $\epsilon = 10^{-6}$. Exponentiating its output gives $\hat{e}$. Both probes use the same optimizer settings as the ensemble. All five neural score routes retain verdict a , including the classifier-confidence route. PhyRoute-D also retains that verdict and scores disagreement before agreement:

$$s_D = \begin{cases} |m_L|, & 1[m_N \geq 0] \neq 1[m_L \geq 0], \\ -|m_L|, & \text{otherwise.} \end{cases} \quad (3)$$

Table 1. Provisional approvals and escalation scores. A score of $-\infty$ makes a point ineligible for fallback. Physics-only and residual routes replace the stale neural verdict before deferral.

Route	Approve	Score
Neural margin	a	$- m_N $
Ensemble spread	a	σ
Spread-normalized margin	a	$- m_N /(\sigma + \epsilon)$
Classifier confidence	a	$- p - 0.5 $
Learned error ratio	a	$\hat{e}/(m_N + \epsilon)$
PhyRoute-D	a	S_D in Eq. (3)
Minimum margin	a	$-\min(m_N , m_L)$
Neural approval-only	a	$-m_L$ if $a = 1$, else $-\infty$
Physics only	b	$- m_L $
Physics approval-only	b	$-m_L$ if $b = 1$, else $-\infty$
Joint approval	$a \wedge b$	$- m_L $
Residual MLP	c	$- m_R $

Always-exact evaluation and exact evaluation whenever topology changes provide reference routes. An exact result replaces the provisional decision on a resolved escalated point.

Primary thresholds are the nominal calibration-score quantiles for 5 %, 10 % and 20 % escalation, then remain fixed. The 10 % setting leads the comparison. Realized test escalation can differ because of shift, ties or approval-only eligibility. Secondary test-score quantiles impose retrospective budgets of 5 %, 10 %, 20 % and 30 %. These diagnose ranking at matched capacity; they are not deployable frozen thresholds. Test labels select neither primary threshold nor score.

For split conformal calibration, define the residual and order-statistic rank

$$r_j = |m_N(x_j) - m(v_j)|, k = \lceil (n_{cal} + 1)(1 - \alpha) \rceil. \quad (4)$$

The interval radius is the k th sorted residual. We use $\alpha = 0.001, 0.01$ and 0.05 , and defer when the margin interval contains zero. Its usual marginal coverage statement requires exchangeability [1]; it does not certify risk after topology shift.

2.4 Verifier mismatch and numerical reference

We run nine verifier conditions in total. All of them use the same physical feeder and injections, but different verifier network models. The first two conditions use the correct current network and the stale original network. The next six conditions use the

current topology with all the resistances and reactances scaled by a factor of 0.8, 0.9, 0.95, 1.05, 1.1 or 1.2. The ninth condition uses independent per-branch resistance/reactance factors in $[0.9, 1.1]$, with one fixed parameter realization for each topology. The residual model is fed with the same perturbed extrema as the other physics-based routes. The thresholds for the nine conditions are the same as those of the nominal original verifier that was calibrated for the nominal original network.

The reference labels are generated using strict backward/forward sweep. Failed batches are subdivided to identify difficult draws, which are then retried using the Newton–Raphson solver provided by pandapower [12]. Only the output that is finite and has converged is labeled. If both methods fail to solve the draw then the row is left unresolved and this does not necessarily mean that the draw is physically infeasible. A recorded amendment was made to handle this case after a predetermined cross-check exposed a failed batch, before comparative evaluation with other methods. The number of generated draws, resolved draws and unresolved draws is recorded for every condition. In paired method tests the same subset of resolved draws is used for comparison.

2.5 Endpoints and computation

To quantify these metrics, we count unsafe approvals per resolved, accepted, and violated test point, as well as false rejections per feasible test point, acceptance coverage, accuracy, and realized escalation. Topology macro averages, worst-topology values and corresponding seed ranges are summarized. Note, however, that these conditional metrics do not bound the risk of the currently unresolved test cases. Repeated predictions on shared operating points cannot justify independent-binomial uncertainty across feeders. Complete-route timing includes inference, routing, and fallback phases, using single-threaded BLAS; uses warm-up and repeated, varied test inputs; records hardware and software details; and reports mean, median, and p95 latency values. Note that reference-label retry costs are not included in the timed workloads above.

3 Results

3.1 Reference coverage and comparison units

The evaluation results for all 549 model–test–condition cells are available. In total, 1,830,000 topology–scenario pairs have been generated, and 1,827,685 of them have been assigned to a converged reference. 2,315 points (out of 30,000 test points for exchange close34_open01) are left unresolved, all of them belonging to this single test case. The unresolved ratio of 0.127% thus hides a numerical problem concentrated to this single test case, where 7.72% of draws remain unresolved.

In the predetermined cross-check with pandapower we reached maximum voltage differences of less than 4×10^{-10} pu among the comparisons in 58 of the 59 exchange batches. The remaining test case required the reference-handling as

described above and is retained in the topology inventory along with the unresolved rows for this case. Nonconvergence of algorithms as well as exclusion from the risk denominator does not imply infeasibility of an operating point.

Unless stated otherwise, the following error rates are computed as macro averages over all 44 topologies in the held-out set, averaged over the nine different combinations of training seeds and test seeds for each topology. Thus, for example, the “unsafe rate” is (unsafe approvals)/(solver-resolved points), and the “false-rejection rate” is (rejected feasible points)/(resolved feasible points), and note that the solver resolves some points and not others, so the unsafe rate and false-rejection rate use different denominators. Escalation uses all generated points. Model evaluations sharing a reference point are repeated predictions on the same physical scenario, not independent physical trials. The seed ranges give a sense of the variability in the results over different random seeds, rather than the confidence interval for a particular estimate.

3.2 Current-physics routes on held-out exchanges

Table 2 reports the frozen 10% calibration setting. The stale neural ensemble without routing has a macro unsafe rate of 17.6519%. Neural-margin routing lowers it to 14.8359%, and spread routing leaves 16.2067%, despite realized escalation near 10%. A classifier-confidence gate, a learned error probe and spread normalization remain in the same high-error range. These results concern models whose operating features omit the changed topology.

Table 2. Nominal current verifier, frozen 10% calibration thresholds. Values are topology-macro percentages on 44 held-out exchanges, averaged over three training and three test seeds. Escalation uses all generated draws; unsafe approvals use resolved draws and false rejections use resolved feasible draws.

Route	Escalation	Unsafe	False rejection
Neural margin	10.22	14.836	2.684
Ensemble spread	9.67	16.207	3.617
Spread-normalized margin	10.18	14.950	2.468
Classifier confidence	10.24	15.126	1.624
Learned error ratio	10.12	14.850	2.477
PhyRoute-D	25.18	0.061	0.079
Minimum margin	12.86	13.409	1.849
Neural approval-only	24.29	0.019	4.489
Physics only	8.66	0.061	0.168
Physics approval-only	8.29	0.023	0.752
Joint approval	8.66	0.060	2.825
Residual MLP	8.40	0.006	0.025

Physics-only routing is a substantially stronger reference. The unsafe approvals and false rejections average 0.06147% and 0.16778% respectively with absolute-margin fallback. The realized escalation is 8.65778%. The residual model improves these averages to 0.00647% and 0.02476% respectively at 8.39836% escalation. The absolute reductions are 0.0550 percentage points for unsafe approvals and 0.1430 points for false rejections. The exact-solve fraction of this learned correction is slightly smaller than that of the linear verifier in the nominal comparison.

The improvement is not uniform. Across training seeds, the residual model’s unsafe rate ranges from 0.00562% to 0.00743%, compared with 0.05779% to 0.06464% for physics-only routing. Its worst-topology unsafe rate is 0.0900%, versus 0.4100%. Paired topology averages show lower unsafe rates on 22 exchanges and lower false-rejection rates on 30. Both errors are no worse on 36 of 44 exchanges. The other eight prevent a claim of dominance on every reconfiguration.

The PhyRoute-D system obtained unsafe approvals of 0.06057% and false rejections of 0.07864% with an escalation of 25.18472% in this case. The improvement of unsafe approvals over the physics-only routing approach is only 0.00090 percentage points while the escalation increases by 16.52694 percentage points. Its lower false-rejection rate may matter under a specified application cost, but the result does not establish an efficient universal gate. The number of feasible points refused cannot be hidden in a single safety metric. The approval-only physics routing approach for example has unsafe approvals of 0.02275% at an escalation of 8.28960% while having false rejections of 0.75225%.

Figure 1 shows the topology means at the frozen 10% calibration setting. Smaller and larger frozen budgets also preserve the nominal ranking. At the frozen 5% calibration setting, the residual route had unsafe approval rate of 0.01843% and false rejection rate of 0.08617% at 4.27977% escalation. At the frozen 20% setting, the residual route had 0.000808% unsafe approvals and 0.001096% false rejections at 16.60066% escalation. On the other hand, the physics-only routing had larger error averages for the corresponding budgets. The error rates shown above are observed tradeoffs over a fixed benchmark, and therefore are not certified risk levels.

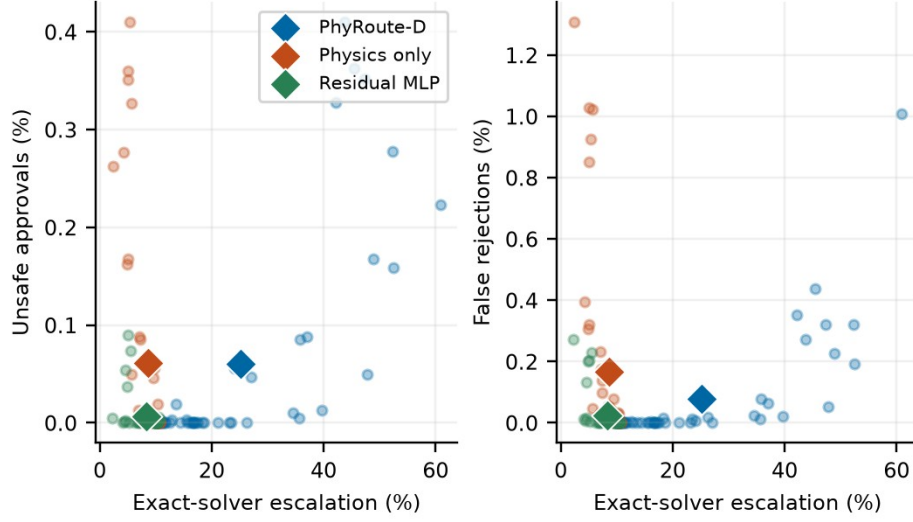


Fig. 1. Three routes with current, correct physics at the frozen 10% calibration setting. Faint points are means for individual held-out topologies; diamonds are the 44-topology macro means. Decision errors use resolved subsets and escalation uses all generated points. Stale neural routes are omitted to preserve the error scale.

3.3 Equal test budgets and nominal calibration

At 10% test escalation (retrospective), the unsafe rate and false-rejection rate for the physics-only routing strategy are 0.01326% and 0.04405%, respectively. For the residual model the corresponding rates are 0.00139% and 0.00105%. Note that the PhyRoute-D strategy has an unsafe rate of 11.40195% and a false-rejection rate of 1.98871%. Its frozen error rate depends on substantially increased escalation. The test-score quantiles give some insight into the ranking capacity of the model, but cannot replace calibration thresholds in a deployment claim.

Note that topology shift is not the same as operating-range shift. On the original topology, all four routing models (neural-margin, physics-only, residual, and disagreement) had zero observed errors at the 10% calibration setting for the nominal operating range. On the photovoltaic-growth test data, the neural-margin routing model had zero observed errors, while the residual routing model had 0.00556% unsafe approvals. Thus, the residual model is not better on all kinds of changes to the data and topology.

The split-conformal route at nominal miscoverage $\alpha=0.001$ has no observed errors on either original-topology distribution, with 3.3011% escalation in the nominal range. On held-out topologies its unsafe rate is 16.6929% with the same 3.3011% escalation. The unchanged rate follows the paired topology-blind inputs. This is a failure of transport to the changed conditional response, not a contradiction of the exchangeability-based conformal statement.

3.4 The supplied verifier can also be wrong

Table 3 and Fig. 2 hold physical truth fixed while changing the verifier. For the residual route, independent branch perturbations in $[0.9, 1.1]$ increase the unsafe rate from 0.00647% to 0.01039%, with 0.04356% false rejections. This single recorded perturbation is milder than a coherent bias and does not estimate performance averaged over possible network errors.

Table 3. Verifier mismatch with frozen nominal 10% calibration thresholds. Entries give unsafe approvals / false rejections in percent, with the same macro averaging and denominators as Table 2. The physical reference is unchanged.

Verifier	PhyRoute-D	Physics only	Residual MLP
Correct	0.061 / 0.079	0.061 / 0.168	0.006 / 0.025
Branch noise $\pm 10\%$	0.107 / 0.104	0.110 / 0.211	0.010 / 0.044
$R, X \times 0.80$	5.193 / 0.007	8.130 / 0.007	7.332 / <0.001
$R, X \times 0.90$	2.308 / 0.021	3.143 / 0.024	2.243 / 0.002
$R, X \times 0.95$	0.918 / 0.040	1.102 / 0.061	0.334 / 0.007
$R, X \times 1.05$	0.013 / 0.161	0.014 / 0.366	0.002 / 0.933
$R, X \times 1.10$	0.005 / 0.517	0.005 / 2.702	0.001 / 5.031
$R, X \times 1.20$	0.002 / 1.814	0.002 / 11.489	<0.001 / 14.046
Stale topology	15.441 / 2.694	15.439 / 2.694	14.860 / 2.631

A coherent factor of 0.9 raises residual-route unsafe approvals to 2.24272%; physics-only routing reaches 3.14304% and PhyRoute-D reaches 2.30788%. The residual model still improves the unsafe average over physics alone in this condition, but its nominal low-error level is lost. At factor 1.1, its unsafe rate falls to 0.001244% while false rejections rise to 5.03135%. Physics-only routing has 0.005498% unsafe approvals and 2.70213% false rejections. Depending on the direction of bias, conservative-looking voltage estimates can exchange missed violations for substantial rejection of feasible points.

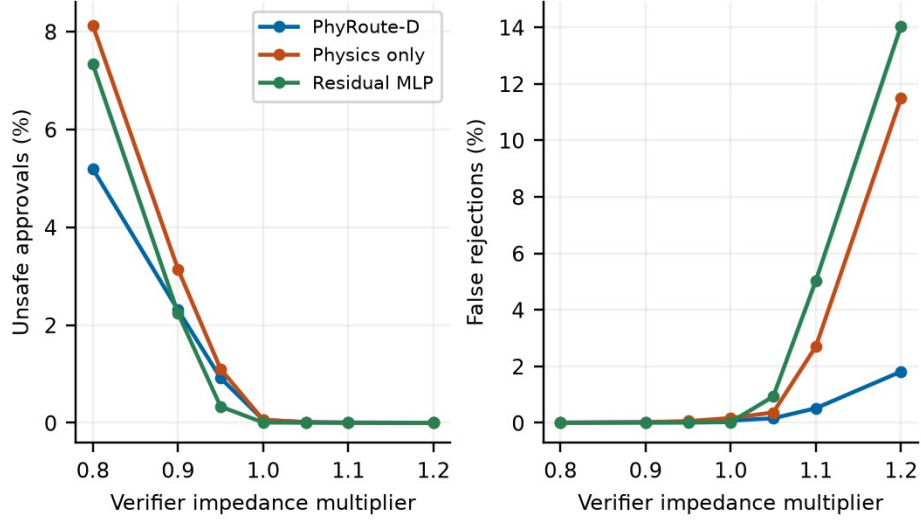


Fig. 2. Sensitivity to coherent error in supplied line impedances, with reference physics and nominal calibration thresholds fixed. Ribbons span the three training-run means, not confidence intervals. Error percentages use resolved denominators. Lower unsafe approvals under overestimated impedance must be read alongside false rejections.

Using the stale original topology removes the useful current-network signal. Residual routing then has 14.85961% unsafe approvals, versus 15.43925% for physics-only routing and 15.44149% for PhyRoute-D. The learned correction does not recover an omitted topology from operating features alone. The nominal result consequently supports a conditional benefit from supplied physical information, not immunity to verifier error.

3.5 Complete-route computation

Table 4 reports complete NumPy routes on an Apple M4 system with Python 3.11.14 and NumPy 2.4.6, with thread counts fixed to one. Timing uses training seed 4101 and test seed 9101, the original topology and the previously explored exchange close35_open31. It covers varied batches of 1, 64 and 512, with two warm-ups, rotating method order and 200, 80 and 30 measured calls, respectively. These are declared resolved workloads with strict backward/forward-sweep fallback, not a measurement of Newton–Raphson retries on difficult reference cases.

Table 4. Measured complete-route mean latency in μs per scenario on an Apple M4 with one BLAS thread. Base is the original network; RC-A is the previously studied close35/open31 exchange. Batch size is shown in each column. Training seed 4101, test seed 9101; frozen 10% calibration thresholds.

Route	Base, 1	RC-A, 1	Base, 512	RC-A, 512
Always exact	396.55	414.72	3.23	3.77
Physics only	64.04	57.81	1.34	1.46
Neural margin	91.88	93.00	1.98	2.13
PhyRoute-D	107.77	130.25	2.10	2.56
Residual MLP	74.45	66.32	1.57	1.72

On the original topology for single-point calls, average time for a physics-only route decision is 64.04 μs , for residual routing is 74.45 μs and for always-exact is 396.55 μs . On the modified topology, the corresponding means are 57.81 μs , 66.32 μs and 414.72 μs respectively. The residual routing median single-point call time for the original topology is 29.85 μs with a p95 of 416.33 μs . In other words, median single-point call latency mostly reflects calls that skip fallback. Reporting median single-point call latency alone greatly underreports the cost of such workloads and hides the large tail of fallbacks.

For a batch size of 512 the mean decision time for single scenarios on the original topology is for the physics-only model 1.339 μs for the residual model 1.570 μs and for the always-exact model 3.226 μs . On the changed topology these mean values rise to 1.459 μs , 1.721 μs and 3.769 μs respectively. For the residual model, more computation buys lower nominal decision errors than the screening of the physics-only model. This higher computation occurs even though the average escalation for the residual model in the held-out study was lower than for the physics-only model. Lower escalation does not automatically equal lower latency. Timing two topologies also cannot establish a latency bound across the inventory or an operational service-level guarantee.

4 Discussion

4.1 Information explains the topology-blind failure

For a fixed trained predictor and routing score g , if the operating-feature distribution is unchanged between two topologies, the distribution of $g(X)$ is unchanged. For any measurable set A ,

$$P_{T_1}\{g(X) \in A\} = \int 1\{g(x) \in A\} dP_X(x) = P_{T_2}\{g(X) \in A\}. \quad (5)$$

This is a direct observation about deterministic functions of the same inputs, not a new theorem. In the paired experiment, the scores agree pointwise across topologies. Labels can still differ because the voltage solution depends on the network. A stable

confidence histogram therefore cannot be interpreted as evidence that the physical decision remains accurate.

LinDistFlow breaks that symmetry by receiving the supplied network. The residual model receives the same physical summary and learns corrections on development exchanges. Its held-out improvement over physics-only screening shows that a learned component can add useful discrimination once current information is supplied. A disagreement gate addresses the stale model differently: it routes its conflicts with physics for resolution. In this benchmark that can be expensive because the stale model itself generates many conflicts. The direct physics and residual routes need not preserve that stale verdict.

4.2 What the comparison establishes

The residual result is an empirical comparison of full systems including many differences with the old ensemble. The residual experiment used 48,000 training data for a number of development topologies, whereas the old ensemble only used 30,000 data points for the original topology. Moreover, the residual experiment used two hidden layers of 64 units each, whereas the old ensemble consisted of 5 networks with 32 units in each of the hidden layers. The input current-physics, the training data diversity, the target function parameterization, the network capacity, and the ensemble overhead all change together in the residual experiment. This makes it impossible to isolate the causal effect of residual learning. Thus, matched ablations are required to make a claim that residual learning alone causes the gain.

Under the stated training and calibration choices, the new route improves nominal error averages over the inexpensive verifier it uses. Runtime and mismatch results indicate the cost of this improved performance and the conditions under which the new route is better. The simple physics-only screening route remains faster and perhaps simpler in the measured routes. When errors do matter, the learned correction offers a quantified alternative. No application-specific cost or acceptable-risk threshold was imposed after observing these outcomes.

However, several significant limitations still apply to the evidence presented here. The exchanges were tested on one single balanced feeder only, so held-out topology does not establish held-out network generalization. The synthetic load and range of photovoltaic power is not meant to represent all possible scenarios. The random impedance test case contains only one realization per topology. Measurement error was not tested. The reference coverage for one of the topologies is still not complete. The revision of this paper followed an exploratory study. The methods and allocation to the different scenarios were, however, ‘frozen’ before the new comparisons. These results do not establish field readiness.

5 Conclusion

Our strongest result is the measured improvement from correcting current LinDistFlow information before applying a calibrated fallback rule. On 44 held-out

branch exchanges, this route reduces both nominal unsafe approvals and false rejections compared to physics-only screening, with some increase in computation but less in terms of escalation than the other two approaches. In contrast, preserving a stale neural verdict through disagreement routing is less efficient than this approach at equal test capacity. However, neither of these approaches provides a general safety certificate: coherent impedance error, a stale topology, and unresolved numerical cases all currently prevent us from drawing that conclusion. Therefore, a credible comparison of how to deploy this must take into account the information each component is given, both decision errors, the realized workload on the solvers and the conditions under which a reference cannot resolve a case.

Data and code. The reproducibility package is available at <https://github.com/Abhinav0905/phyroute>. Software release v0.2.0 is archived at <https://doi.org/10.5281/zenodo.22905467>. It contains the protocol and amendment, topology allocation, configuration and source hashes, non-executable model arrays, reference-coverage records, evaluation tables and plotting and timing code. Large scenario and physics arrays are regenerable with the supplied serial pipeline; their metadata and hashes are retained in the archive. The original exploratory experiment is preserved separately.

AI assistance. ChatGPT assisted with initial idea development. Claude developed the original experimental code and manuscript draft. Codex assisted with code and numerical audits, protocol revision, literature verification, revised experiment tooling, drafting and typesetting. Writeless.ai assisted with language revision, followed by a technical audit and correction of the revised text. Human authors retain responsibility for validating the evidence, interpreting the findings and approving the submitted text.

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